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%% System Parameters (Protein Dynamics - Example 3)
A_p = [0.7 0; 0 0.52];
S_p = [0.18 0; 0 0.14];
R_p = [0.4 0; 0 0.3];
Ko = 0; Kq = 1; alpha = 0.01;

% Paper Results for Protein Dynamics
P_p = [0.9815, 0; 0, 0.9806];
J_p = [0.8130, 0; 0, 0.8219];
E_p = [0.5164, 0; 0, 0.5670];

%% Simulation Loop
k_max = 100;
xp = zeros(2, k_max);
xp(:,1) = [0.6; 0.2]; % Initial protein levels

for k = 1:k_max-1
    if k <= 50
        wk = [0.5*cos(k); 1.8*sin(k)];
    else
        wk = [0; 0];
    end

    % Dynamics for proteins (xp)
    yk = A_p * xp(:,k) + R_p * wk;
    fk = max(min(yk, 1), Ko); % Nonlinearity f(y)
    xp(:,k+1) = fk + S_p * wk;
end

%% Plotting Protein Trajectories
figure('Color', 'w');
plot(0:k_max-1, xp(1,:), 'b', 'LineWidth', 1.5); hold on;
plot(0:k_max-1, xp(2,:), 'c', 'LineWidth', 1.5);
grid on;
line([50 50], ylim, 'Color', 'r', 'LineStyle', '--');
title('Example 3 (Part 2): Protein Expression Levels');
xlabel('k'); ylabel('Concentration x_p(k)');
legend('Protein Alpha', 'Protein Beta');

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